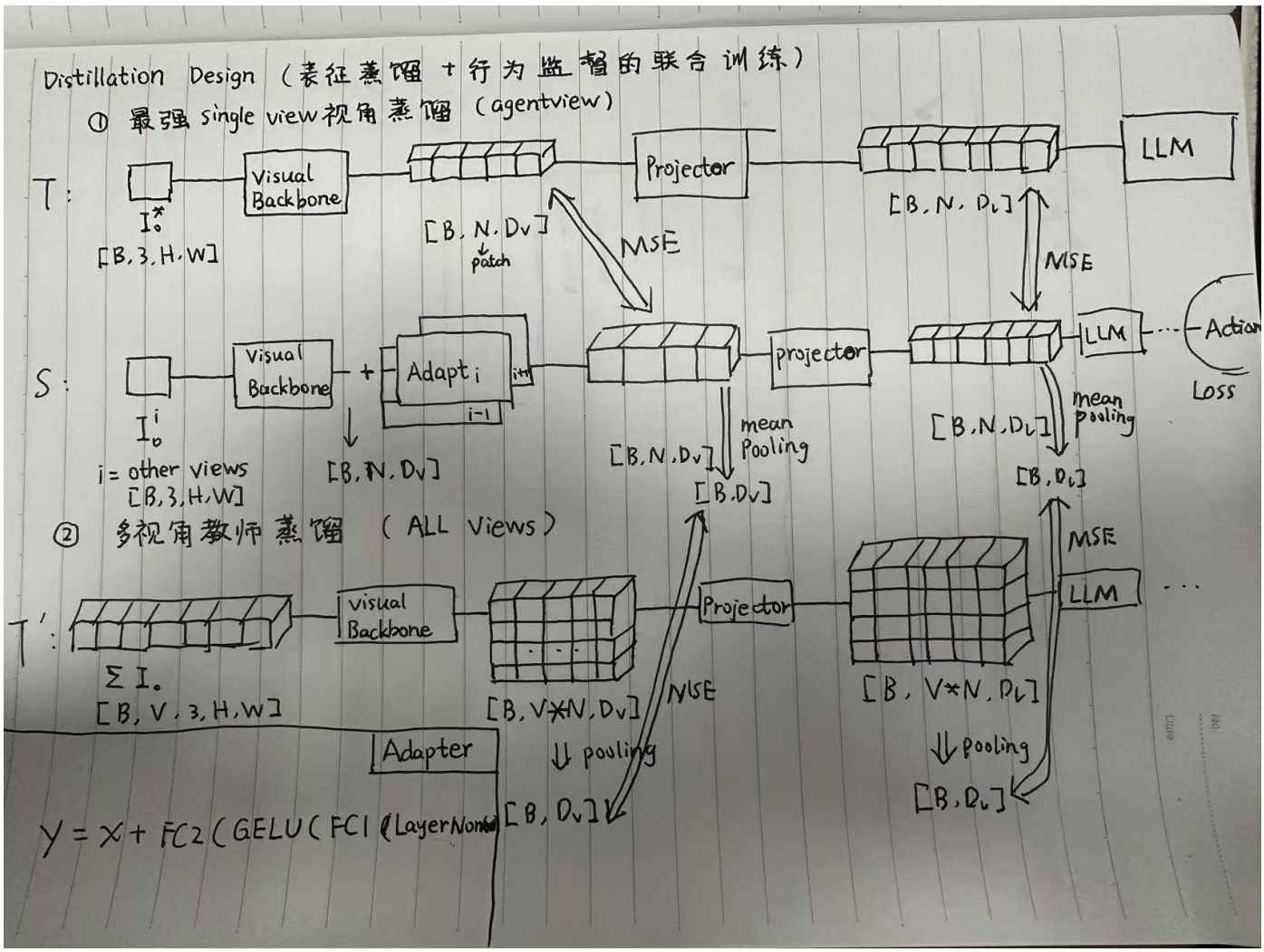
多视角蒸馏训练

蒸馏多视角网络架构设计

Agentview 40k libero spatial 微调ckpt为teacher；蒸馏其他任意视角输入。

蒸馏网络设计：



a. 其中teacher的patch级视觉表征与教师的visual backbone后输出+不同视角adapter输出的patch级表征做MSE。
b. projector后的表征做mse
c. Action动作测进行行为监督

单视角训练消融

batchsize32 vs batchsize16

模型	Task 1	Task 2	Task3	Task 4	Task 5	Task 6	Task 7	Task 8	Task 9	Task 10	ALL
	pick up the black bowl between the plate and the ramekin and place it on the plate	pick up the black bowl next to the ramekin and place it on the plate	pick up the black bowl from table center and place it on the plate	pick up the black bowl on the cookie box and place it on the plate	pick up the black bowl in the top drawer of the wooden cabinet and place it on the plate	pick up the black bowl on the ramekin and place it on the plate	pick up the black bowl next to the cookie box and place it on the plate	pick up the black bowl on the stove and place it on the plate	pick up the black bowl next to the plate and place it on the plate	pick up the black bowl on the wooden cabinet and place it on the plate	
Agentview bs16 scheduler off 40k libero spatial benchmark	96	88	98	96	80	20	98	90	78	66	81
Agentview bs32 scheduler off 40k libero spatial benchmark	90	82	96	98	60	18	98	90	78	66	77.6

结论

在scheduler off的情况下，bs16训练40k效果更好

是否开启动态学习率linear-warmup+cosine-decay

模型	Task 1	Task 2	Task 3	Task 4	Task 5	Task 6	Task 7	Task 8	Task 9	Task 10	ALL
	pick up the black bowl between the plate and the ramekin and place it on the plate	pick up the black bowl next to the ramekin and place it on the plate	pick up the black bowl from table center and place it on the plate	pick up the black bowl on the cookie box and place it on the plate	pick up the black bowl in the top drawer of the wooden cabinet and place it on the plate	pick up the black bowl on the ramekin and place it on the plate	pick up the black bowl next to the cookie box and place it on the plate	pick up the black bowl on the stove and place it on the plate	pick up the black bowl next to the plate and place it on the plate	pick up the black bowl on the wooden cabinet and place it on the plate	
Agentview bs32 scheduler on 40k libero spatial benchmark	92	90	100	90	64	20	96	88	88	72	80
Agentview bs32 scheduler off	90	82	96	98	60	18	98	90	78	66	77.6

40k libero spatial benchmark											

结论

1. 在bs32的情况下，scheduler on在40k训练效果更好。
2. 结合在scheduler off的情况下，bs16训练40k效果更好：猜测最好的训练是bs16，scheduler on，40k
3. 需要补充实验：agentview，bs16，scheduler-on，40k，libero spatial benchmark

模型	Task 1	Task 2	Task 3	Task 4	Task 5	Task 6	Task 7	Task 8	Task 9	Task 10	ALL
	pick up the black bowl between the plate and the ramekin and place it on the plate	pick up the black bowl next to the ramekin and place it on the plate	pick up the black bowl from table center and place it on the plate	pick up the black bowl on the cookie box and place it on the plate	pick up the black bowl in the top drawer of the wooden cabinet and place it on the plate	pick up the black bowl on the ramekin and place it on the plate	pick up the black bowl next to the cookie box and place it on the plate	pick up the black bowl on the stove and place it on the plate	pick up the black bowl next to the plate and place it on the plate	pick up the black bowl on the wooden cabinet and place it on the plate	
Agentview bs16 scheduler on 40k libero spatial benchmark	90	90	94	92	86	8	96	88	86	58	78.8

Agentview bs16 scheduler off 40k libero spatial benchmark	96%	88%	98%	96%	80%	20%	98%	90%	78%	66%	81%

结论

和设想不同，agentview，bs16，40k，libero spatial benchmark使用scheduler on效果不如scheduler off

更新认知：

a. 在bs32时，scheduler on能提高模型验证成功率
b. 在bs16时，scheduler on会降低模型验证成功率
c. 后续实验打算统一用32k，因此需要开启scheduler on

不同视角模型能力验证

模型	Task 1	Task 2	Task3	Task 4	Task 5	Task 6	Task 7	Task 8	Task 9	Task 10	ALL
	pick up the black bowl between the plate and the ramekin and place it on	pick up the black bowl next to the ramekin and place it on the plate	pick up the black bowl from table center and place it on the plate	pick up the black bowl on the cookie box and place it on the plate	pick up the black bowl in the top drawer of the wooden cabinet and place	pick up the black bowl on the ramekin and place it on the plate	pick up the black bowl next to the cookie box and place it on the plate	pick up the black bowl on the stove and place it on the plate	pick up the black bowl next to the plate and place it on the plate	pick up the black bowl on the wooden cabinet and place it on the plate	

	the plate				it on the plate						
Agentview bs16 scheduler off 40k libero spatial benchmark	96%	88%	98%	96%	80%	20%	98%	90%	78%	66%	81%
Topview bs16 scheduler off 40k libero spatial benchmark	66%	96%	88%	90%	60%	34%	50%	94%	76%	80%	73.4 %
Leftview bs16 scheduler off 40k libero spatial benchmark	62%	78%	52%	54%	40%	12%	62%	62%	64%	54%	54%
Rightview bs16 scheduler off 40k libero spatial benchmark	62%	82%	78%	66%	46%	12%	76%	76%	64%	34%	59.6 %
Backview	38%	62%	16%	62%	56%	12%	54%	26%	40%	40%	

bs16 scheduler off 40k libero spatial benchmark											40.6%
Leftbackview bs16 scheduler off 40k libero spatial benchmark	88%	84%	86%	96%	70%	14%	88%	88%	46%	66%	72.6%
Rightbackview bs16 scheduler off 40k libero spatial benchmark	88%	76%	60%	56%	54%	36%	72%	70%	68%	36%	61.6%

结论

Agentview > topview > leftbackview > rightbackview > rightview > leftview > backview

多视角训练消融

验证不同多视角能力以及是否比单视角强

模型	Task 1	Task 2	Task 3	Task 4	Task 5	Task 6	Task 7	Task 8	Task 9	Task 10	ALL

	pick up the black bowl between the plate and the ramekin and place it on the plate	pick up the black bowl next to the ramekin and place it on the plate	pick up the black bowl from table center and place it on the plate	pick up the black bowl on the cookie box and place it on the plate	pick up the black bowl in the top drawer of the wooden cabinet and place it on the plate	pick up the black bowl on the ramekin and place it on the plate	pick up the black bowl next to the cookie box and place it on the plate	pick up the black bowl on the stove and place it on the plate	pick up the black bowl next to the plate and place it on the plate	pick up the black bowl on the wooden cabinet and place it on the plate	
2view (agentview+topview) bs32 scheduler on 40k libero spatial benchmark	96	94	94	94	80	6	94	90	88	78	81.4
4view (agentview+topview+leftbackview+rightbackview) bs32 scheduler on 40k libero spatial benchmark	92	92	96	94	88	18	98	90	82	64	81.4

7view (agentview+topview +leftview+rightview+ backview+leftbackvi ew+rightbackview) bs32 scheduler on 40k libero spatial benchmark											
Agentview bs32 scheduler on 40k libero spatial benchmark	92	90	100	90	64	20	96	88	88	72	80

验证多视角batchsize影响

模型	Task 1	Task 2	Task 3	Task 4	Task 5	Task 6	Task 7	Task 8	Task 9	Task 10	ALL
	pick up the blac k bowl betw een the plat e and the ram ekin and plac	pick up the blac k bowl next to the ram ekin and plac e it on the	pick up the blac k bowl from tabl e cent er and plac e it on the	pick up the blac k bowl on the cook ie box and plac e it on the plat e	pick up the blac k bowl in the top draw er of the woo den cabi net and plac	pick up the blac k bowl on the ram ekin and plac e it on the plat e	pick up the blac k bowl next to the cook ie box and plac e it on the	pick up the blac k bowl on the stov e and plac e it on the plat e	pick up the blac k bowl next to the plat e and plac e it on the	pick up the blac k bowl on the woo den cabi net and plac e it on the	

	e it on the plate	plate	plate		e it on the plate		plate		plate	plate	
7view (agentview+topview+leftview+rightview+backview+leftbackview+rightbackview) bs16 scheduler on 40k libero spatial benchmark	96%	88%	88%	98%	64%	24%	100%	90%	76%	70%	79.4%
7view (agentview+topview+leftview+rightview+backview+leftbackview+rightbackview) bs32 scheduler on 40k libero spatial benchmark											

学生蒸馏训练消融

> 备注：学生蒸馏有2种训练模式。
>
> 1. 加上所有的adapter，但是数据采样和参数更新只对特定视角生效。用着种方法可以训练出单视角的蒸馏模型，耗时更少，但训练完成的模型只能用于特定视角验证；
>
> 2. 加上所有adapter，数据采样均匀且随机视角，根据训练数据更新对应的adapter参数。需要耗费更长的训练时间，但是训练完成后的蒸馏模型可以用于任意视角的验证。

Baseline:

1. Single view finetuned
    a. backview

b. bs32
c. scheduler on
d. 40k
e. libero spatial benchmark

2. Student distillation
   a. Backview
   b. bs32
   c. Scheduler on
   d. 40k
   e. Action loss 1
   f. Patch loss 0
   g. Projector 0
   h. libero spatial benchmark

验证蒸馏学生模型training steps影响

模型	Task 1	Task 2	Task 3	Task 4	Task 5	Task 6	Task 7	Task 8	Task 9	Task 10	ALL	
	pick up the black bowl between the plate and the ramekin and place it	pick up the black bowl next to the ramekin and place it on the plate	pick up the black bowl from table center and place it on the	pick up the black bowl on the cookie box and place it on the plate	pick up the black bowl in the top drawer of the wooden cabinet and place it	pick up the black bowl on the ramekin and place it on the plate	pick up the black bowl next to the cookie box and place it on the	pick up the black bowl on the stove and place it on the plate	pick up the black bowl next to the plate and place it on the plate	pick up the black bowl on the wooden cabinet and place it on the plate		

	on the plate		plate		on the plate		plate					
Backview finetuned a. bs32 b. scheduler on c. 40k d. libero spatial benchmark	48	64	38	52	42	12	54	22	54	50	43.6	
40k agentview teacher bs32 scheduler on libero spatial benchmark 40k backview student bs32 scheduler on Action loss 1.0 Patch loss 1.0	64	48	48	52	52	18	54	60	40	52	48.8	+5.2(11.9%)

Projector loss 1.0 libero spatial benchmark												
40k agentview teacher bs32 scheduler on libero spatial benchmark 50k backview student bs32 scheduler on Action loss 1.0 Patch loss 1.0 Projector loss 1.0 libero spatial benchmark	54	66	70	62	40	14	64	60	46	52	52.8	+9.2(21.1%)

验证蒸馏学生模型能否比单视角微调效果好

模型											ALL	

	Task 1	Task 2	Task 3	Task 4	Task 5	Task 6	Task 7	Task 8	Task 9	Task 10		
	pick up the black bowl between the plate and the ramekin and place it on the plate	pick up the black bowl next to the ramekin and place it on the plate	pick up the black bowl from table center and place it on the plate	pick up the black bowl on the cookie box and place it on the plate	pick up the black bowl in the top drawer of the wooden cabinet and place it on the plate	pick up the black bowl on the ramekin and place it on the plate	pick up the black bowl next to the cookie box and place it on the plate	pick up the black bowl on the stove and place it on the plate	pick up the black bowl next to the plate and place it on the plate	pick up the black bowl on the wooden cabinet and place it on the plate		
Backview finetuned a. bs32 b. scheduler on c. 40k d. libero spatial benchmark	48	64	38	52	42	12	54	22	54	50	43.6	
40k agentview	64	48	48	52	52	18	54	60	40	52	48.8	+5.2(11.9

teacher bs32 scheduler on libero spatial benchmark 40k backview student bs32 scheduler on Action loss 1.0 Patch loss 1.0 Projector loss 1.0 libero spatial benchmark												%)
Topview finetuned a. bs32 b. scheduler on c. 40k d. libero spatial benchmark	76	90	88	90	60	16	76	96	68	64	72.4	

40k agentview teacher bs32 scheduler on libero spatial benchmark 40k topview student bs32 scheduler on Action loss 1.0 Patch loss 1.0 Projector loss 1.0 libero spatial benchmark	54	86	74	80	36	22	76	70	58	54	61	-11.4 (15.7%)

验证Loss，adapter模块设计的有效性

模型	Task 1	Task 2	Task 3	Task 4	Task 5	Task 6	Task 7	Task 8	Task 9	Task 10	ALL
	pick up the black bowl	pick up the black bowl	pick up the black bowl	pick up the black bowl	pick up the black bowl	pick up the black bowl	pick up the black bowl	pick up the black bowl	pick up the black bowl	pick up the black bowl	

	between the plate and the ramekin and place it on the plate	next to the ramekin and place it on the plate	from table center and place it on the plate	on the cookie box and place it on the plate	in the top drawer of the wooden cabinet and place it on the plate	on the ramekin and place it on the plate	next to the cookie box and place it on the plate	on the stove and place it on the plate	next to the plate and place it on the plate	on the wooden cabinet and place it on the plate	
Backview finetuned a. bs32 b. scheduler on c. 40k d. libero spatial benchmark	48	64	38	52	42	12	54	22	54	50	43.6
Student distillation a. Backview b. bs32 c. Scheduler on d. 40k e. Action loss 1 f. Patch loss 0 g. Projector 0	62	78	58	50	58	12	58	26	50	48	50

h. libero spatial benchmark											
Student distillation a. Backview b. bs32 c. Scheduler on d. 40k e. Action loss 1 f. Patch loss 1 g. Projector 0 h. libero spatial benchmark	52	78	68	70	58	6	36	50	46	52	51.6
Student distillation a. Backview b. bs32 c. Scheduler on d. 40k e. Action loss 1 f. Patch loss 0 g. Projector 1 h. libero spatial benchmark	62	56	54	58	56	6	70	36	56	42	49.6

Student distillation a. Backview b. bs32 c. Scheduler on d. 40k e. Action loss 1 f. Patch loss 1 g. Projector 1 h. libero spatial benchmark	64	48	48	52	52	18	54	60	40	52	48.8

结论

Projector loss对模型eval成功率有副作用

验证单视角蒸馏和多视角同时蒸馏对模型效果的影响

> 备注：学生蒸馏有2种训练模式。
>
> 1. 加上所有的adapter，但是数据采样和参数更新只对特定视角生效。用着种方法可以训练出单视角的蒸馏模型，耗时更少，但训练完成的模型只能用于特定视角验证；
>
> 2. 加上所有adapter，数据采样均匀且随机视角，根据训练数据更新对应的adapter参数。需要耗费更长的训练时间，但是训练完成后的蒸馏模型可以用于任意视角的验证。

模型	Task 1	Task 2	Task 3	Task 4	Task 5	Task 6	Task 7	Task 8	Task 9	Task 10	ALL
	pick up the black bowl	pick up the black bowl next	pick up the black bowl	pick up the black bowl on	pick up the black bowl in	pick up the black bowl on	pick up the black bowl next	pick up the black bowl on	pick up the black bowl next	pick up the black bowl on	

	between the plate and the ramekin and place it on the plate	to the ramekin and place it on the plate	from table center and place it on the plate	the cookie box and place it on the plate	the top drawer of the wooden cabinet and place it on the plate	the ramekin and place it on the plate	to the cookie box and place it on the plate	the stove and place it on the plate	to the plate and place it on the plate	the wooden cabinet and place it on the plate	
Student distillation(all view) a. Backview eval b. bs32 c. Scheduler on d. 40k * 6 views = 240k e. Action loss 1 f. Patch loss 1 g. Projector 1 h. libero spatial benchmark											
Student distillation(single view) a. Backview eval	64	48	48	52	52	18	54	60	40	52	48.8

b. bs32 c. Scheduler on d. 40k e. Action loss 1 f. Patch loss 1 g. Projector 1 h. libero spatial benchmark											